IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

The project followed an end-to-end data science workflow:

Data Collection: Gathered launch data using the SpaceX API and web scraping.

Data Wrangling: Cleaned and transformed data for analysis.

SQL Analysis: Explored launch trends and mission performance.

Data Visualization: Used charts and interactive analytics to identify patterns.

Machine Learning: Built predictive models such as Logistic Regression, SVM, Decision Tree, and KNN to predict landing success.

Geospatial Analysis: Visualized launch sites using interactive maps.

Summary of all results

The results showed that factors such as payload mass, orbit type, launch site, and booster version influence landing success. Machine learning models achieved effective prediction performance, demonstrating that historical launch data can be used to estimate future landing outcomes. The project successfully provided insights into Falcon 9 mission performance and reusable rocket success patterns.

Introduction

Project background and context

SpaceX has transformed the aerospace industry by developing reusable rockets, particularly the Falcon 9 launch vehicle. One of the key cost-saving strategies used by SpaceX is the successful recovery and reuse of the first-stage booster after launch. Since booster recovery significantly reduces launch expenses, predicting landing success has become an important problem for mission planning and cost estimation.

This project aims to answer the following key questions:

- Can we predict the success of Falcon 9 first-stage landings using historical launch data?
- Which factors most influence landing success, such as payload mass, launch site, orbit type, and booster version?
- Do certain launch sites have higher landing success rates than others?
- How does payload mass impact the probability of successful landing?
- Which machine learning model performs best for predicting landing outcomes?
- Can interactive dashboards and visualizations help better understand launch performance patterns?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - **SpaceX REST API:** Used to obtain historical Falcon 9 launch information, including payload mass, launch site, orbit type, booster version, and landing outcomes.
  - **Web Scraping:** Additional launch information was extracted from publicly available web pages using web scraping techniques.
- Perform data wrangling
  - Removing irrelevant information
  - Handling missing values
  - Cleaning inconsistent data
  - Converting data into proper formats
  - Creating a target variable for landing success (1 = successful landing, 0 = unsuccessful landing)

Methodology

- Perform exploratory data analysis (EDA) using visualization and SQL
- Using SQL:
  - Analyzed launch frequencies
  - Examined launch site performance
  - Compared mission success rates
  - Investigated payload characteristics
- Using Visualization:
  - Scatter plots to analyze payload vs landing success
  - Bar charts for launch site performance
  - Correlation analysis for feature relationships

Methodology

- Perform interactive visual analytics using Folium and Plotly Dash
- **Folium:**
  - Interactive maps were created to visualize:
  - Falcon 9 launch sites
  - Successful and failed launches
  - Geographic patterns of launch activities
- **Plotly Dash:**
  - An interactive dashboard was developed allowing users to:
  - Filter launches by site
  - Select payload ranges
  - Explore landing success visually through charts and graphs

Methodology

Perform predictive analysis using classification models:

- Logistic Regression
- SVM
- Decision Tree
- KNN

How to build, tune, evaluate classification models:

- Selected key features (payload, orbit, launch site)
- Split data into train/test sets
- Trained multiple models
- GridSearchCV for hyperparameter tuning
- Evaluated using accuracy and model comparison
- Selected best performing model based on test results

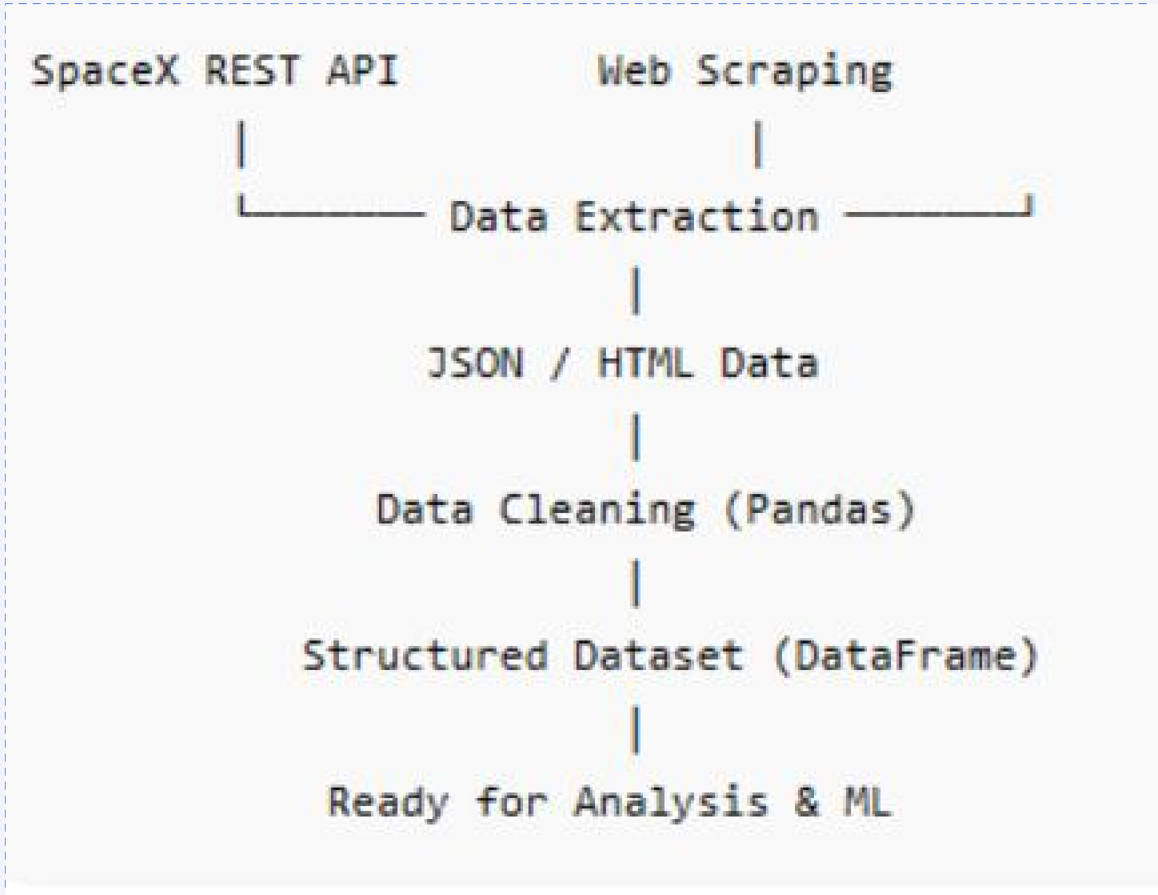
Data Collection

Describe how data sets were collected.

- Data collected from SpaceX REST API
- Historical Falcon 9 launch records retrieved in JSON format
- Additional launch information gathered using web scraping techniques
- Data extracted included: launch site, payload mass, orbit type, booster version, landing outcome
- Raw data converted into structured format using Python (Pandas)
- Final dataset prepared for analysis and modeling

Data Collection – SpaceX API

- https://github.com/luqman-abban/IBM-Data-Science-Capstone/blob/main/01_Data_Collection_API.ipynb



Data Collection – Scraping

- Data extracted from public SpaceX-related web pages
- Used BeautifulSoup for HTML parsing
- Sent HTTP requests using Requests library
- Targeted structured elements (tables, tags, and attributes)
- Extracted launch details not fully available via API
- Cleaned and standardized scraped data
- Converted results into Pandas DataFrame for analysis
- https://github.com/luqman-abban/IBM-Data-Science-Capstone/blob/main/02_Web_Scraping.ipynb

Target Web Pages

|
▼

HTTP Request (Requests Library)

|
▼

HTML Content Received

|
▼

Parsing with BeautifulSoup

|
▼

Extract Required Data (Tables / Tags)

|
▼

Data Cleaning & Formatting (Pandas)

|
▼

Structured Dataset Ready for Analysis

Data Wrangling

Data was processed using following:

- Loaded raw datasets into Pandas DataFrames
- Identified and handled missing values
- Removed irrelevant and duplicate records
- Standardized data formats (dates, numeric fields, categories)
- Converted categorical variables into numerical form
- Created target variable: Landing Outcome (0 = Fail, 1 = Success)
- Performed feature selection for modeling
- Prepared final clean dataset for analysis and machine learning

- https://github.com/luqman-abban/IBM-Data-Science-Capstone/blob/main/03_Data_Wrangling.ipynb

```
Raw Data (API + Web Scraping)
        |
        ▼
Load into Pandas DataFrame
        |
        ▼
Data Inspection (missing values, types, duplicates)
        |
        ▼
Data Cleaning
- Remove duplicates
- Handle missing values
- Fix inconsistencies
        |
        ▼
Data Transformation
- Standardize formats
- Encode categorical variables
- Create new features
        |
        ▼
Feature Selection
        |
        ▼
Final Clean Dataset
(Ready for EDA & Machine Learning)
```

EDA with Data Visualization

Following charts were used for EDA Visual Analytics:

- Flight Number vs Payload Mass (Scatter Plot): Showed that higher flight experience improves success, while heavier payloads reduce success rate.
- Flight Number vs Launch Site (Scatter Plot): Compared success trends across different launch sites.
- Payload Mass vs Launch Site (Scatter Plot): Showed payload distribution differences across sites.
- Orbit vs Success Rate (Bar Chart): Identified which orbits have higher landing success rates.
- Flight Number vs Orbit (Scatter Plot): Showed improvement in success over time for some orbits.
- Payload vs Orbit (Scatter Plot): Showed how payload affects success in different orbit types.
- Yearly Success Trend (Line Chart): Showed increasing success rate over time, especially after 2013.
- https://github.com/luqman-abban/IBM-Data-Science-Capstone/blob/main/05_Interactive_Visual_Analytics.ipynb

EDA with SQL

The SQL queries were performed for the following:

- Extracted unique launch sites from the SpaceX dataset
- Filtered records where Launch_Site starts with "CCA" and limited results
- Calculated total payload mass for NASA (CRS) missions
- Computed average payload mass for booster version F9 v1.1
- Found earliest successful ground pad landing date using MIN()
- Listed boosters with drone ship success and payload mass between 4000–6000 kg
- Counted mission outcomes grouped by success/failure categories
- Identified boosters carrying maximum payload using a subquery with MAX()
- Extracted 2015 failure (drone ship) records with month parsing via substr()
- Ranked landing outcomes by frequency within a specific date range (2010–2017)

https://github.com/luqman-abban/IBM-Data-Science-Capstone/blob/main/04_SQL_Analysis.ipynb

Build an Interactive Map with Folium

- Added circle markers and labeled points to show each SpaceX launch site on the map for easy identification.
- Plotted success (green) and failure (red) markers for individual launches and grouped them using MarkerCluster to avoid clutter.
- Enabled MousePosition tool to inspect exact latitude/longitude while exploring nearby features.
- Used DivIcon text markers to display calculated distances (in km) from launch sites to nearby points like coastline, city, railway, and highway.
- Drew PolyLine connections between launch sites and these nearby features to visually show spatial relationships.

Overall purpose: to visualize launch locations, mission outcomes, and their proximity to key geographic and infrastructure features in an interactive way.

https://github.com/luqman-abban/IBM-Data-Science-Capstone/blob/main/07_Folium_Locations_Analysis.ipynb

Build a Dashboard with Plotly Dash

- This interactive dashboard analyzes SpaceX launch performance using site selection, payload, and launch outcome data.
- **Visualizations Included**
  - 1. Pie Chart – Launch Success Distribution: Shows success vs failure of launches. Compares performance across launch sites (ALL sites view). Also shows outcomes for a selected site
  - 2. Scatter Plot – Payload vs Launch Outcome: X-axis: Payload Mass (kg). Y-axis: Launch outcome (Success = 1, Failure = 0)
- **Purpose of Dashboard**
  - Identify most successful launch sites
  - Understand how payload affects success
  - Compare booster performance

Build a Dashboard with Plotly Dash

Interactive Features of the Dashboard

- **Launch Site Dropdown**
  - Select ALL sites or a specific site
  - Dynamically updates both charts
- Payload Range Slider
  - Filters data based on payload weight
  - Helps analyze performance across light and heavy payloads 1. Data Preparation

https://github.com/luqman-abban/IBM-Data-Science-Capstone/blob/main/08_Dashboard_App.ipynb

Predictive Analysis (Classification)

Loaded SpaceX dataset

- Defined target: Class (landing success/failure)
- Standardized features using StandardScaler
- Split data into 80% training / 20% testing

Model Building

- Trained 4 classification models:
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)

Optimization

- Used GridSearchCV (10-fold cross-validation)
- Tuned hyperparameters for each model to find best settings

Predictive Analysis (Classification)

Evaluation

- Measured performance using test accuracy + confusion matrix
- Compared all models on unseen data

Results

- Logistic Regression: ~0.83
- SVM: ~0.83
- KNN: ~0.83
- Decision Tree: ~0.67

Best Model

Best performers: Logistic Regression, SVM, KNN (tie)

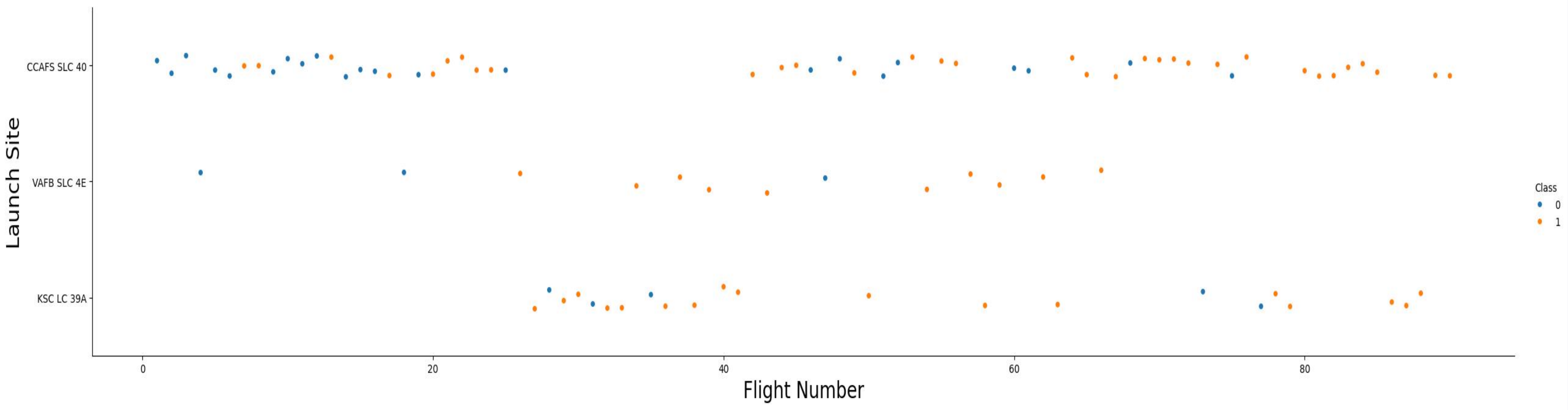
Decision Tree performed worst due to lower generalization

```
START
  ↓
Load Dataset (SpaceX Falcon 9)
  ↓
Feature Selection (X, Y)
  ↓
Data Standardization (StandardScaler)
  ↓
Train-Test Split (80/20)
  ↓
-------------------------------------
|  MODEL BUILDING PHASE             |
|-----------------------------------|
| Logistic Regression               |
| SVM                               |
| Decision Tree                     |
| KNN                               |
-------------------------------------
  ↓
Hyperparameter Tuning (GridSearchCV, 10-fold CV)
  ↓
Model Training (Best Params Selected)
  ↓
Model Evaluation:
   - Cross-validation Accuracy
   - Test Accuracy
   - Confusion Matrix
  ↓
Compare Model Performance
  ↓
Select Best Model
  ↓
FINAL OUTPUT: Best Landing Prediction Model
  ↓
END
```

Section 2

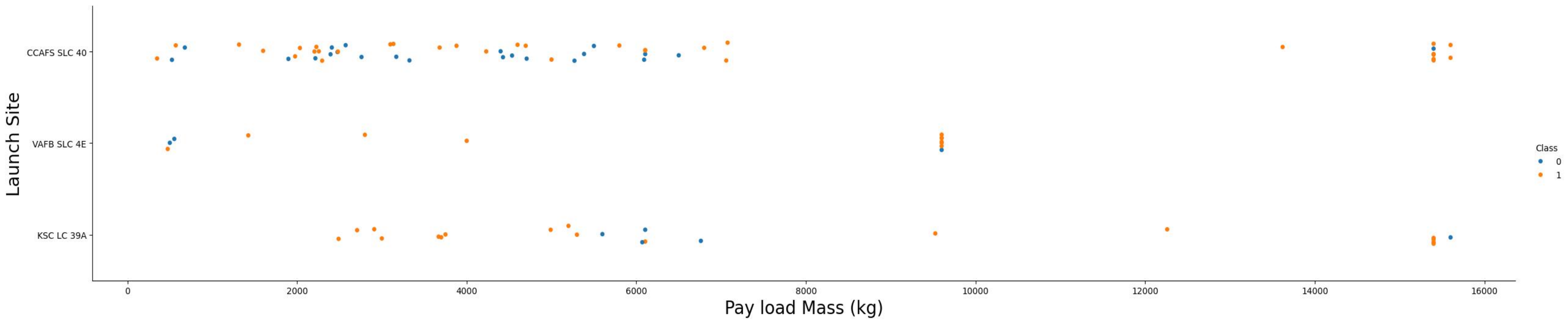
Insights drawn from EDA

Flight Number vs. Launch Site



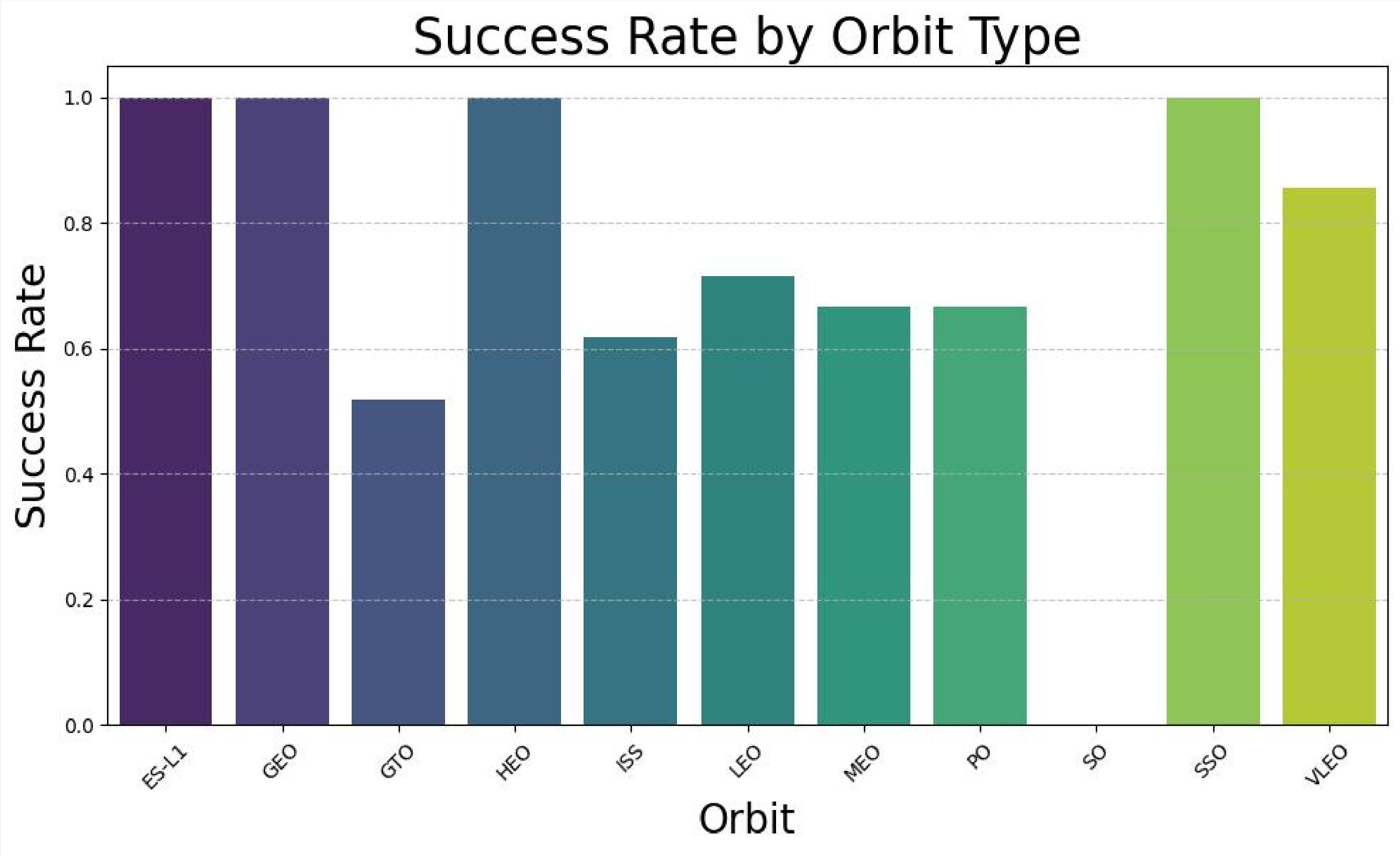
- CCAFS SLC 40 is the most active site, with launches spread across all flight numbers (1–100+), showing consistent use throughout SpaceX's history
- KSC LC 39A became active around flight ~30 and has seen steady use since, particularly for later missions
- VAFB SLC 4E has the fewest launches and appears used selectively, likely for polar/sun-synchronous orbits

Payload vs. Launch Site

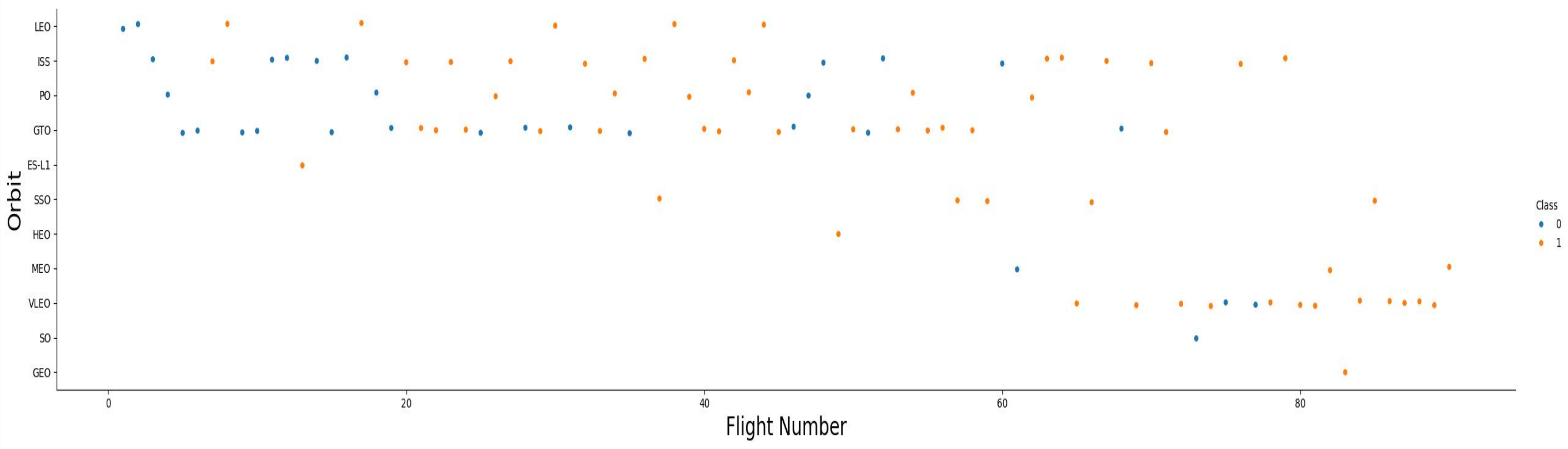


- CCAFS SLC 40 handles the widest range of payload masses (~0–15,500 kg), confirming its role as SpaceX's most versatile and frequently used pad
- KSC LC 39A tends to handle mid-to-high payload masses (~2,000–15,500 kg), consistent with its use for heavier or crewed missions
- VAFB SLC 4E is clustered at lower masses (under ~1,500 kg) with a notable exception near ~9,500 kg, reflecting its typical use for lighter polar orbit payloads

Success Rate vs. Orbit Type



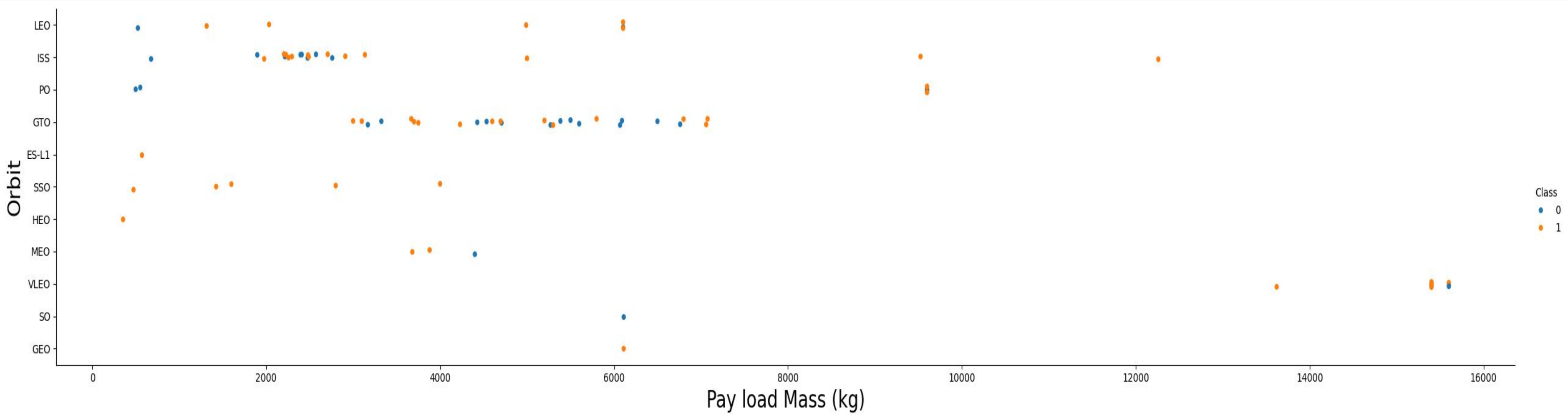
Flight Number vs. Orbit Type



- ISS & LEO dominate throughout — these are the most common mission targets
- Class 0 (blue) is early; Class 1 (orange) grows later — suggesting a shift in mission outcomes or rocket type over time
- Orbit diversity increases with higher flight numbers — missions became more varied and ambitious over time
- VLEO is a late addition (flight 70+), while GTO fades after flight 70

In short: early flights were simpler and concentrated on ISS/LEO, while later flights became more diverse in orbit type with a higher proportion of Class 1 outcomes.

Payload vs. Orbit Type

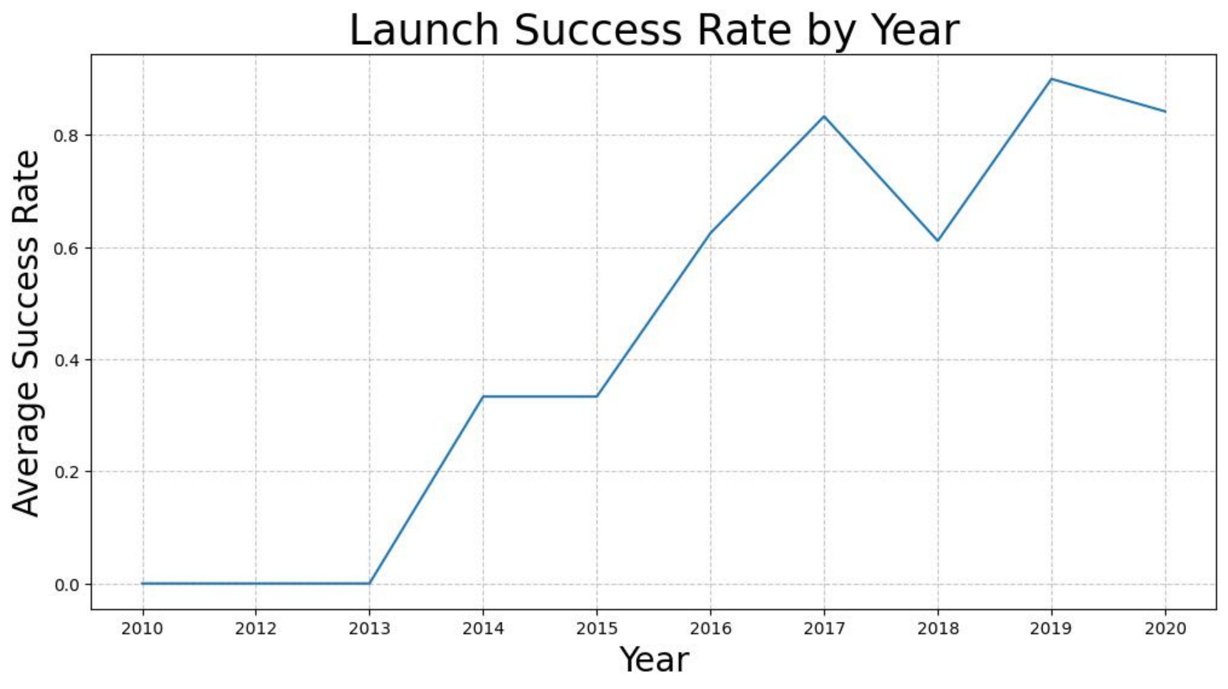


- The plot compares payload mass, orbit type, and launch success/failure.
- ISS, LEO, and GTO have the most launches.
- Successful landings (Class 1) are more frequent overall.
- Both successes and failures occur at all payload sizes, so payload mass alone does not predict success.
- VLEO and ISS missions show relatively higher success rates.
- Some orbits (HEO, GEO, SSO) have very few launches.
- Overall, orbit type may influence success more than payload mass.

Launch Success Yearly Trend

- 2010–2012: Success rate was 0% — all early launches failed or the metric wasn't being tracked
- 2013–2015: Gradual improvement, rising from near 0 to ~35%
- 2015–2017: Sharp acceleration, jumping to ~83% — a major reliability breakthrough
- 2018: Notable dip to ~61% — a setback year
- 2019: Peak performance at ~90% — best year on record
- 2020: Slight decline to ~85%, but still strong

Overall trend: A clear upward trajectory from complete failure to ~85–90% success, with 2018 being the only significant setback in an otherwise strong improvement curve.



All Launch Site Names

- Following are the unique Launch Site names extracted with SQL query:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

A table of SpaceX's launche sites beggining with 'CCA' (2010–2013), showing:

- All missions were successful, mostly carrying Dragon capsules to ISS/LEO
- All boosters were early Falcon 9 v1.0 versions
- Landing recovery failed or wasn't attempted every time — booster reuse was not yet viable

Total Payload Mass

- 45,596 kg is the total payload mass delivered by SpaceX on behalf of NASA

Total_Payload_Mass
45596

Average Payload Mass by F9 v1.1

The Falcon 9 v1.1 booster carried an average payload of ~2.9 metric tons per mission — a modest capacity, reflecting its role as an early-generation rocket before SpaceX significantly upgraded payload performance in later versions.

Average_Payload_Mass
2928.4

First Successful Ground Landing Date

- First Successful Ground Pad Landing: December 22, 2015
- This was a historic milestone — the first-ever successful recovery of a Falcon 9 booster on a ground pad, proving that orbital-class rockets could be landed and reused, a key step in SpaceX's mission to revolutionize spaceflight economics.

Successful Drone Ship Landing with Payload between 4000 and 6000

- All four are Falcon 9 Full Thrust variants. The .2 suffix on B1021 and B1031 indicates these were reused/reflown boosters — proving that recovered rockets could successfully land on drone ships even when carrying heavier payloads.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Out of 101 total missions, SpaceX achieved a ~99% success rate — with only 1 in-flight failure on record, demonstrating exceptional mission reliability across their Falcon 9 program.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters that Carried Maximum Payload

All 12 boosters are Falcon 9 Block 5 (B5) variants — SpaceX's most advanced and final Falcon 9 version.

Key observations:

- All are reflown boosters (indicated by .2 through .7 suffixes), meaning reused rockets carried the heaviest loads
- B1049 and B1051 were flown the most times (up to 7 and 6 times respectively)
- This confirms that Block 5 + reused boosters represent SpaceX's peak payload capability

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Both failures were early v1.1 boosters launching from the same site (CCAFS LC-40) in the first half of 2015 — this was SpaceX's experimental phase of drone ship landings, before achieving their first successful drone ship landing in April 2016.

Failed Drone Ship Landings in 2015:

Month	Booster	Launch Site
January	F9 v1.1 B1012	CCAFS LC-40
April	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

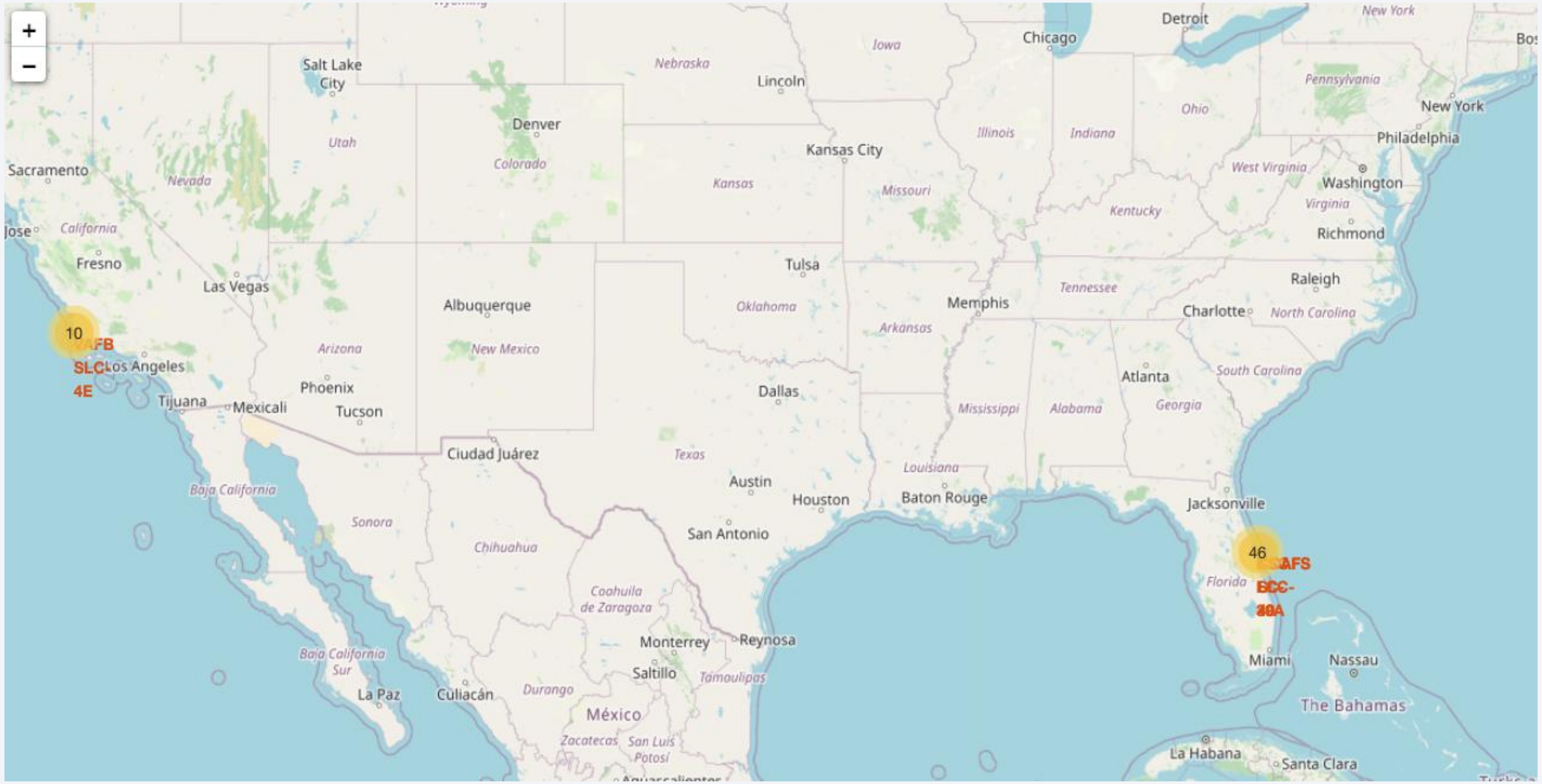
- No attempt dominates — early missions didn't even try landing recovery
- Drone ship had an equal 5 successes vs 5 failures — still being mastered during this period
- Ground pad was more reliable (3 successes, 0 failures)
- Overall, this period reflects SpaceX's trial-and-error phase of booster recovery

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

SpaceX Falcon 9 Launch Sites & Mission Frequency Map



SpaceX Falcon 9 Launch Sites & Mission Frequency Map

Two Launch Clusters:

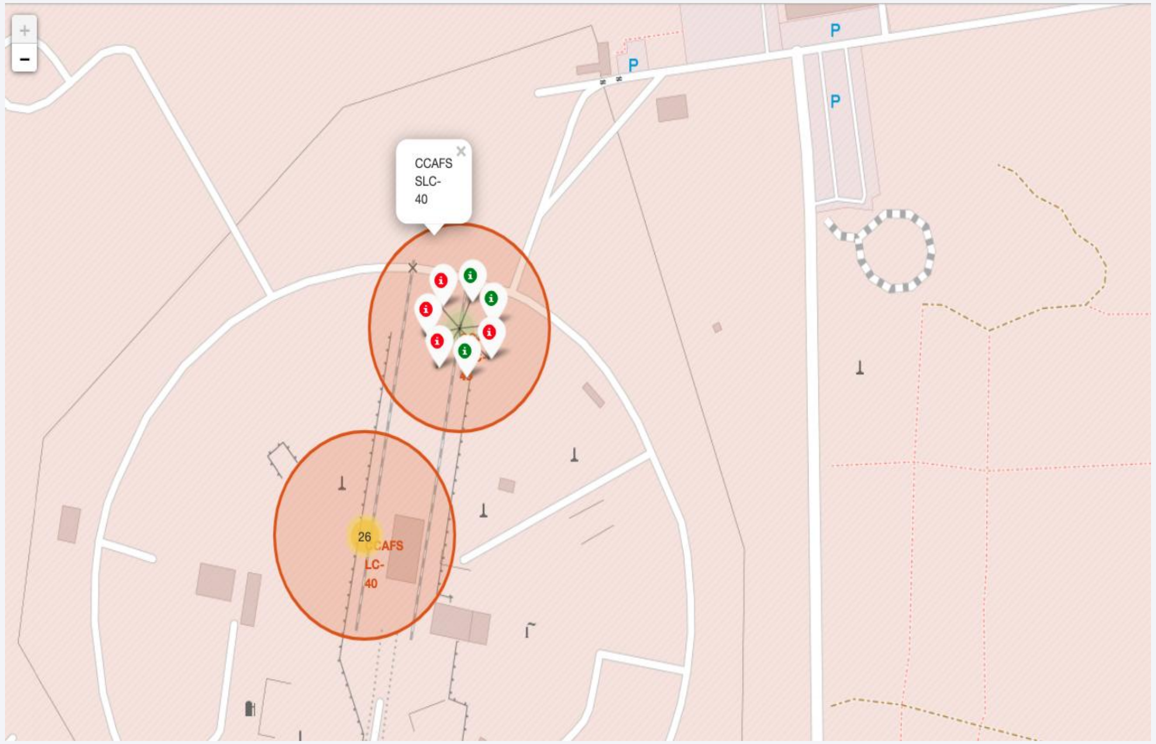
- West Coast — Vandenberg (California)
  - Sites: VAFB SLC-4E (Vandenberg Air Force Base)
  - 10 launches — smaller cluster
  - Used primarily for polar & SSO orbits
- East Coast — Cape Canaveral (Florida)
  - Sites: CCAFS LC-40 and KSC LC-39A
  - 46 launches — dominant cluster
  - Used for ISS, LEO, and GTO missions

Key Findings:

- Florida dominates with ~82% of all launches — ideal for ISS and equatorial orbits
- California handles specialized missions requiring high-inclination orbits
- Both sites are coastal — essential for safe downrange flight paths over the ocean
- The large size difference in clusters (46 vs 10) reflects CCAFS/KSC's role as SpaceX's primary launch hub

CCAFS LC-40 — Launch Outcome Map by Mission

- Red markers = Failed landing outcomes
- Green markers = Successful landing outcomes
- Orange circles = Proximity/cluster zones around the launch pad
- "26" yellow bubble = 26 total launches from CCAFS LC-40
- Popup label = "CCAFS SLC-40" identifying the active pad



Findings:

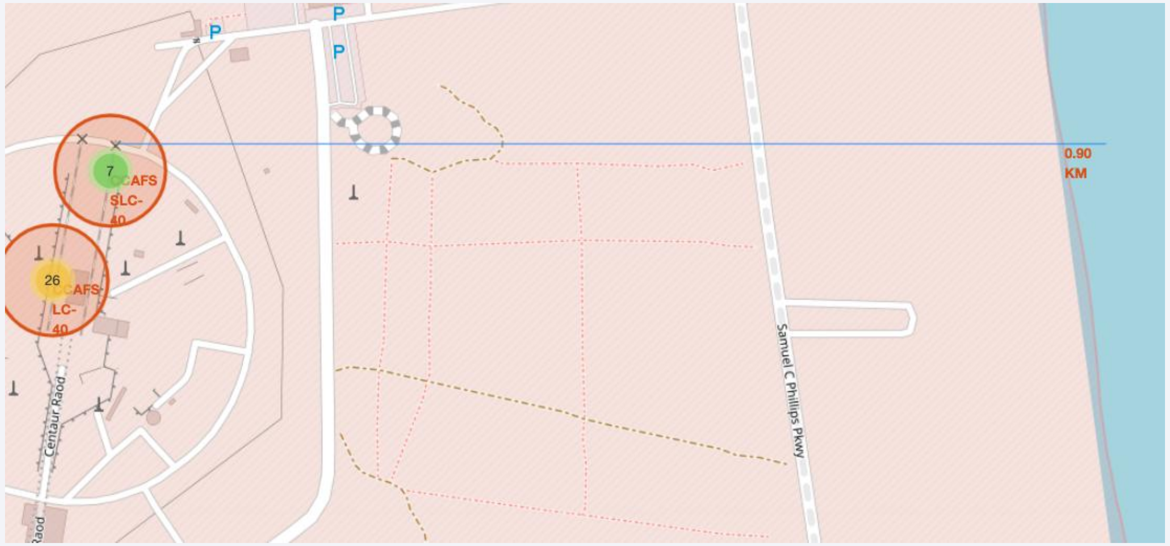
- 26 missions from CCAFS LC-40 — SpaceX's busiest pad
- Mixed success/failure landings — still in experimental phase
- Two zones visible — likely main pad + secondary area

CCAFS Launch Site Proximity Map — Distance to Coastline

- Yellow bubble (26) = CCAFS LC-40 — 26 total launches
- Green bubble (7) = CCAFS SLC-40 — 7 launches
- Blue horizontal line = Distance measurement to the coastline
- 0.90 KM = Distance from the launch site to the Atlantic coastline
- Orange circles = Launch site boundary zones

Findings:

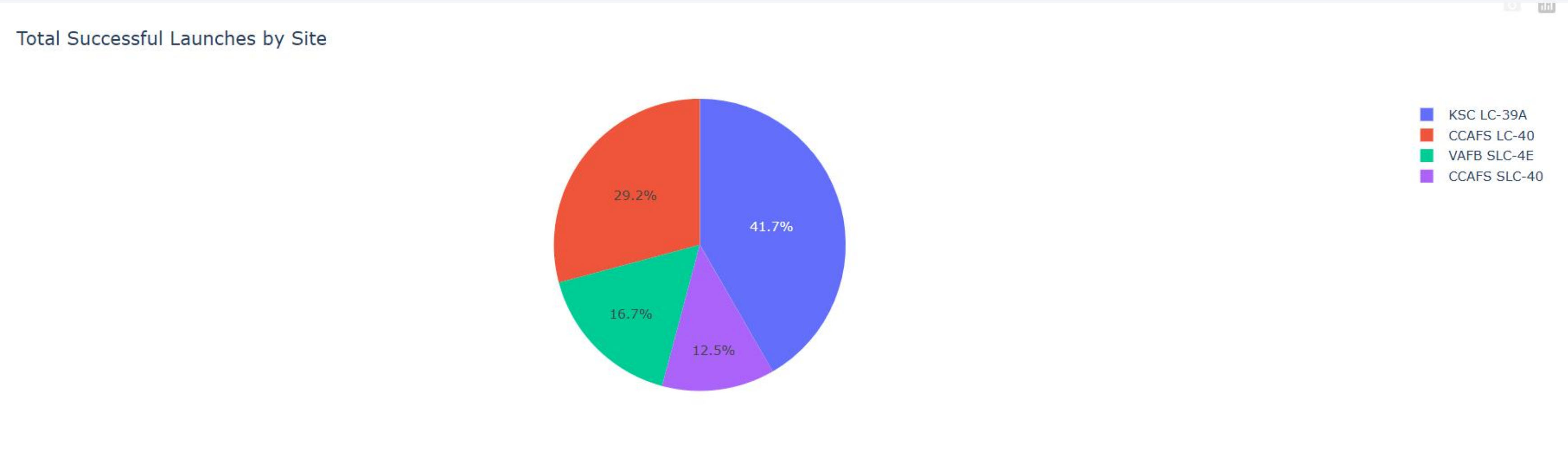
- The launch pad sits just 0.90 km from the coastline — intentionally close for safe over-ocean trajectories
- Road access (Samuel C. Phillips Pkwy) provides direct infrastructure support to the pad
- The two pads are extremely close to each other, sharing the same facility zone
- Coastal proximity also aids booster recovery via drone ships deployed offshore



Section 4

Build a Dashboard with Plotly Dash

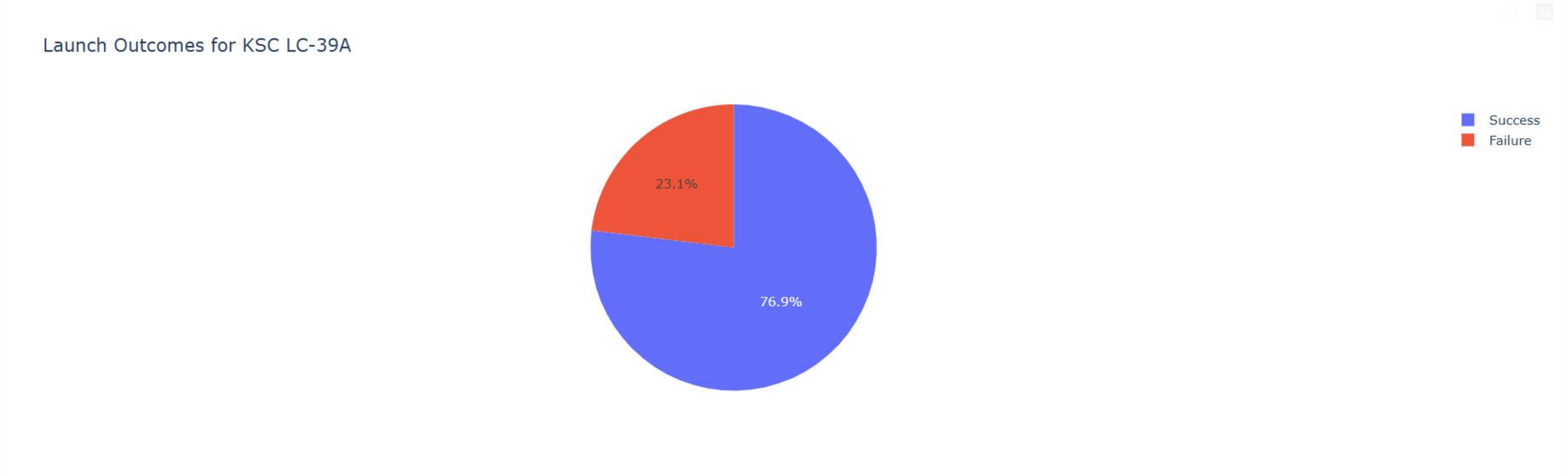
Total Successful Launches by Site — Summary



Key Findings:

- KSC LC-39A dominates with 41.7% — SpaceX's most successful single pad, historically NASA's Apollo & Shuttle pad
- Florida sites combined (KSC + CCAFS) account for ~83% of all successful launches
- VAFB SLC-4E (California) contributes 16.7% — handling specialized polar orbit missions
- CCAFS SLC-40 is the least used at 12.5%

Launch Outcomes for KSC LC-39A



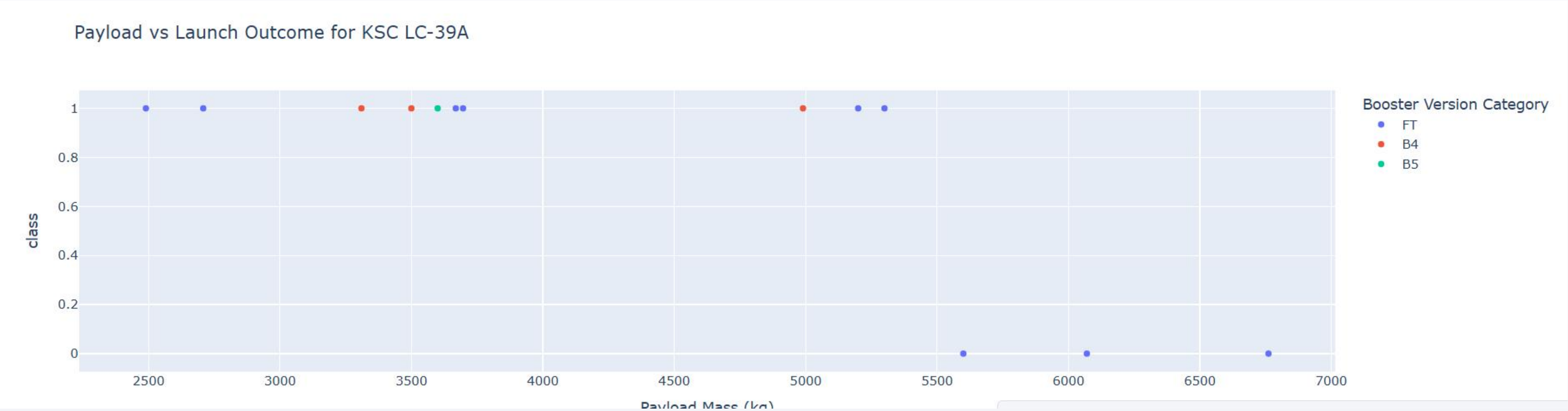
Key Findings:

- KSC LC-39A has the highest success ratio among all SpaceX launch sites at 76.9%
- Nearly 3 out of every 4 launches from this pad succeeded
- The 23.1% failure rate likely includes early booster landing failures, not mission failures
- KSC LC-39A is SpaceX's most reliable and most used pad — a fitting legacy for a site that also launched Apollo and Space Shuttle mission

Payload vs Launch Outcome for KSC LC-39A

Key Findings:

- All failures (class 0) occur at heavier payloads (5,500–6,500 kg) — suggesting higher mass increases risk
- FT boosters dominate this site but show failures at the heavier end
- B4 and B5 both succeeded at mid-range payloads (~3,300–3,600 kg)
- Light-to-mid payloads (2,400–5,000 kg) show a near-perfect success rate
- B5 only appears once — at ~3,600 kg with a successful outcome



Section 5

Predictive Analysis (Classification)

Classification Accuracy

Key takeaway — Logistic Regression wins (tied but preferred):

Model	Validation	Test
Logistic Regression	84.6%	83.3%
SVM	84.8%	83.3%
KNN	84.8%	83.3%
Decision Tree	87.5%	66.7%

Weakest model
Decision Tree

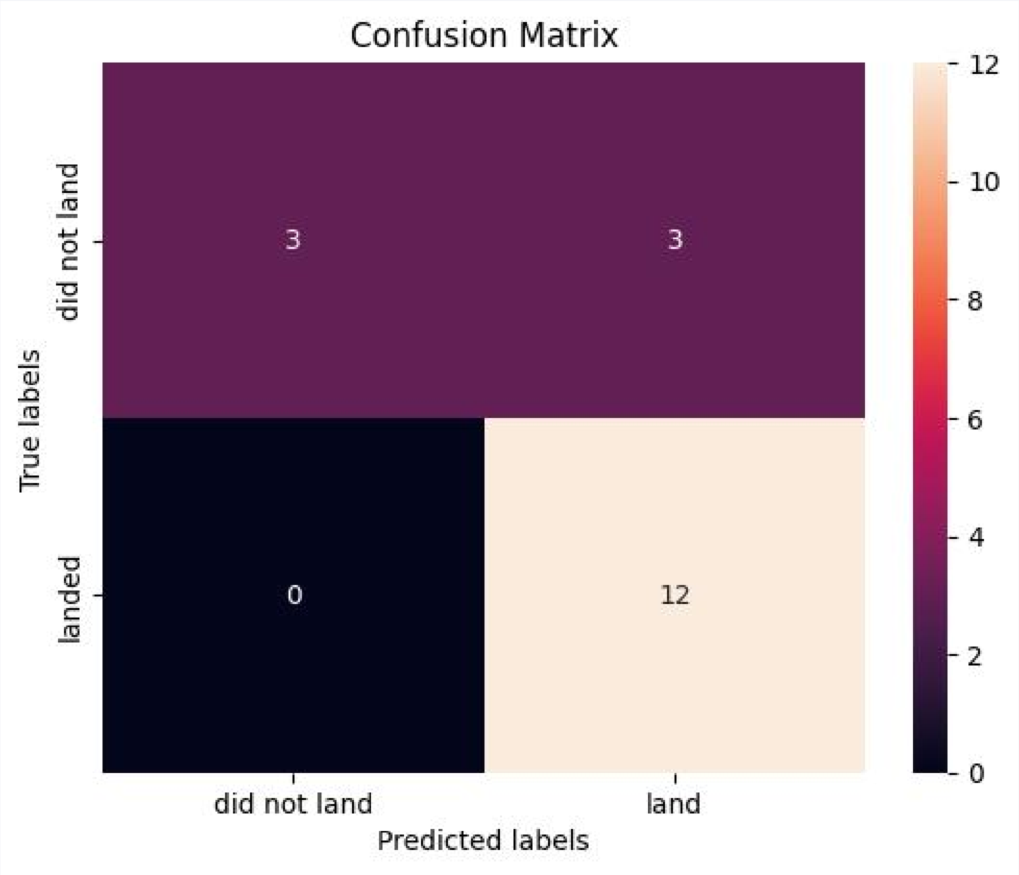
Top test accuracy
83.3%

Best model
Logistic Regression

- Three models (LR, SVM, KNN) tie at 83.3% test accuracy. Logistic Regression is selected as the best because it achieves the same test accuracy with the simplest model — less prone to overfitting. The Decision Tree, despite the highest validation score (87.5%), drops sharply on test data, revealing significant overfitting.

Confusion Matrix

- 15/18 correct predictions — 83.
- 15/18 correct predictions — 83.3% accuracy
- 12 true positives — all actual landings correctly predicted
- 3 true negatives — correctly identified non-landings
- 0 false negatives — never missed a real landing
- 3 false positives — only weakness; occasionally over-predicts landings



Conclusions

Mission Success

- SpaceX achieved ~99% mission success across 101 Falcon 9 flights
- Success rate grew from 0% (2010) to 90% (2019) — a remarkable decade of improvement

Booster Recovery

- First successful ground pad landing: Dec 22, 2015 — proving reusable rockets are viable
- Block 5 reused boosters are the most reliable and capable, carrying the heaviest payloads
- Drone ship landings were mastered over time — equal failures and successes pre-2017, improving after

Launch Sites

- KSC LC-39A is the best performing site — 76.9% success rate, 41.7% of all successful launches
- Florida dominates (~83% of launches); California handles niche polar orbit missions

Conclusions

Key Factors Affecting Landing Success

- Heavier payloads (5,500+ kg) increase failure risk
- Orbit type matters — ISS and VLEO show higher success rates
- Later booster versions (Block 5) significantly outperform earlier ones

Predictive Modeling

- Landing success can be predicted with 83.3% accuracy using historical data
- Logistic Regression is the best model — simple, reliable, zero missed landings
- Decision Tree overfit and is not recommended for this use case

Thank you!